



UNSW
SYDNEY

COMP4141

Theory of Computation

Lecture 5: Proving Non-Regularity

Outline

Pumping Lemma

Myhill-Nerode Theorem

Summary of Regular Languages

Appendix: Binary Relations

Appendix: Proof of Myhill-Nerode Theorem

How to Prove Non-Regularity

Finite automata have their limits when it comes to characterising languages. A DFA with k states must have visited at least one of its states repeatedly to accept a word of length $> k$. It has gone through a loop. The DFA must therefore also accept all words generated by going through that loop any number of times. This is what we call *pumping*.

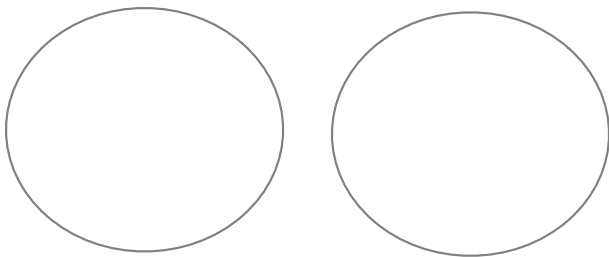
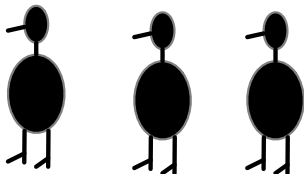
The Pumping Lemma

Theorem (Pumping Lemma)

If $L \subseteq \Sigma^$ is regular then there exists $p \in \mathbb{N}$ (the pumping length) where, if $w \in L$ with $|w| \geq p$, then w may be split into three pieces, $w = xyz$, satisfying the following conditions:*

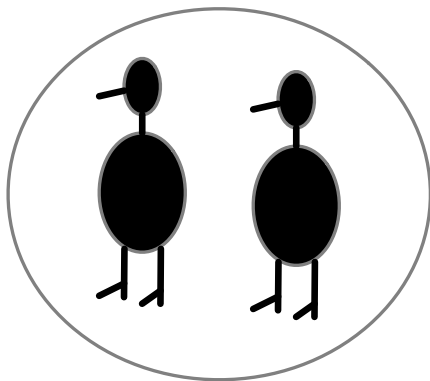
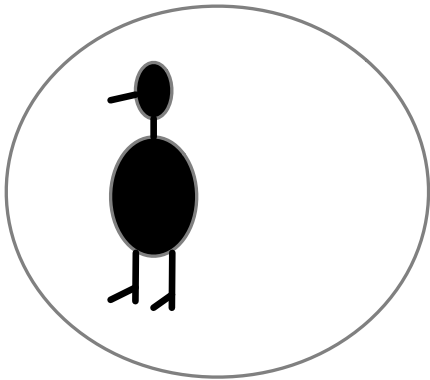
- 1 $|xy| \leq p$,
- 2 $|y| > 0$, and
- 3 $xy^iz \in L$, for all $i \in \mathbb{N}$.

The Pigeon Hole Principle



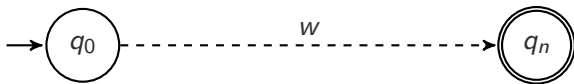
Suppose some number of pigeons are headed for a rest in some number of holes.

The Pigeon Hole Principle



If there are more pigeons than holes, then some hole will have more than one pigeon in it.

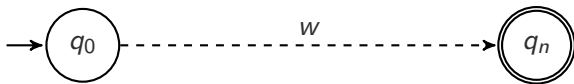
Proof of the pumping Lemma



$$n \geq |Q|$$

If there are more steps in a run of a finite state automaton than the automaton has states

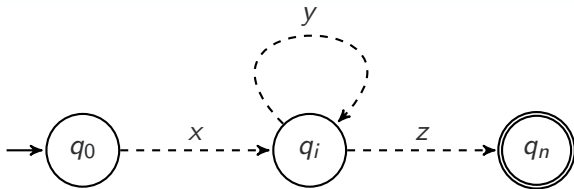
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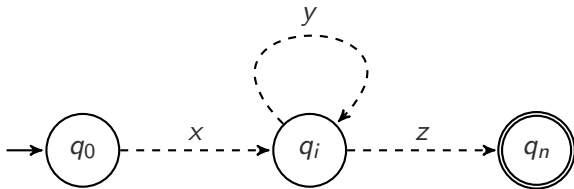


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Take q_i to be the *first* repeated state.

Proof of the pumping Lemma



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If there are more steps in a run of a finite state automaton than the automaton has states then some state occurs more than once in the run.

Take q_i to be the *first* repeated state.

Then

- $|xy| \leq p = |Q|$
- $|y| > 0$ and
- $xy^kz \in L(A)$ for all $k \geq 0$.

Non-Regularity Proofs with the Pumping Lemma

How can we use the Pumping Lemma to prove that a given language L is not regular?

Non-Regularity Proofs with the Pumping Lemma

How can we use the Pumping Lemma to prove that a given language L is not regular?

We use its contrapositive, that is:

$$L \text{ can't be pumped} \Rightarrow L \text{ is not regular.}$$

Non-Regularity Proof Example

Example

Consider $L = \{ a^i b^i \mid i \in \mathbb{N} \}$.

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Case $y = \epsilon$: violates 2.

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Choose $w = a^p b^p$.

Next we show that **regardless of how we split w into xyz** , none of these splits satisfies all three conditions given in the Pumping Lemma.

Case y contains b 's: Then $|xy| > p$, violating condition 1.

Case $y = \epsilon$: violates 2.

Case y consists only of a 's: Then $xyyz$ contains more a s than b s and is thus not in L_1 , violating condition 3.

Non-Regularity Proof Example

Example

Show that $L = \{0^{2^n} : n \in \mathbb{N}\}$ is not regular.

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The Myhill-Nerode Theorem

This theorem provides another exact characterisation of the regular languages. We use it to prove non-regularity—mostly when the pumping lemma fails.

Let $L \subseteq \Sigma^*$. If there exists a $z \in \Sigma^*$ such that $xz \in L$ and $yz \notin L$ (or vice versa), we call x and y *distinguishable by L* .

We write $x \equiv_L y$ if x and y are not distinguishable by L .

(Equivalently, $x \equiv_L y$ when $xz \in L \Leftrightarrow yz \in L$ for all $z \in \Sigma^*$.)

NB

$\equiv_L$ is an equivalence relation on Σ^* .

▶ what?

The Myhill-Nerode Theorem

We write $[w]_L$ for the $\equiv_L$ -equivalence class $\{ v \in \Sigma^* \mid w \equiv_L v \}$ of w .

Definition

The *index of L* is the number of its $\equiv_L$ -equivalence classes.

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Definition

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Theorem (Myhill-Nerode)

$L \subseteq \Sigma^*$ is regular iff the index of L is finite. Moreover, the index is the minimum number of states of a DFA that accepts L .

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Sketch of proof.

“ $\Rightarrow$ ”: Let $A = (Q, \Sigma, \delta, q_0, F)$ be a DFA with $L(A) = L$. Show that $|Q| \geq \text{index of } L$.

“ $\Leftarrow$ ”: Assume the index of L is finite. Define the DFA

$$A_L = (\{ [w]_L \mid w \in \Sigma^* \}, \Sigma, \delta_L, [\epsilon]_L, F_L)$$

where $\delta_L([w]_L, a) = [wa]_L$ and $F_L = \{ [w]_L \mid w \in L \}$. Show that $L(A_L) = L$.

[▶ details](#)



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We use the contrapositive of the “ $\Rightarrow$ ” direction of the theorem.

More specifically, it suffices to find an infinite sequence $u_0, u_1, \dots$ of strings such that for all $i \neq j$ there exists a string w_{ij} such that $u_i w_{ij} \in L$ but $u_j w_{ij} \notin L$ (or vice versa).

It follows that $u_i \notin [u_j]_L$ for all distinct $i, j \in \mathbb{N}$ and hence there are infinitely many $\equiv_L$ -equivalence classes, that is, the index of L is not finite.

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Example

- $L = \{ a^i b^i \mid i \in \mathbb{N} \}$. Choose $u_i = a^i$ and $w_{ij} = b^i$.
- $L = \{ 0^{2^n} : n \in \mathbb{N} \}$. Choose $u_i = w_{ij} = 0^{2^i}$

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Regular Languages

Definition

A language L is **regular** if $L = L(A)$ for some DFA A .

Alternative characterizations:

- Language accepted by an NFA
- Language accepted by an ϵ -NFA
- Language matched by a regular expression
- Language with a finite Myhill-Nerode index

Closure Properties

Regular languages are closed under:

- Determinism/Non-determinism
- Complement
- Intersection
- Union, Set difference, Symmetric difference
- Regular operations: Union, Concatenation, Kleene star
- $\text{odd}(\cdot)$ and $\text{rev}(\cdot)$
- Cone operations: Homomorphisms, Inverse homomorphisms, Intersection
- ...

Limitations

Regular languages are limited to:

- Finite memory calculations:
 - Count up to a fixed number
 - Counting in a fixed modulus (e.g. counting parity)
 - Storing a fixed amount of information
- Finite input strings
- Yes/No outputs (i.e. set membership)

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Reminder: Binary Relations

Let A and B be sets. Any $R \subseteq A \times B$ is called a *binary relation*. We often write aRb for $(a, b) \in R$.

If the first and second set are the same, i.e. $R \subseteq A^2$, we call R a *binary relation on A* and define the following properties of such relations. Say that R is:

- *reflexive* if aRa for all $a \in A$.
- *irreflexive* if aRa for no $a \in A$.
- *transitive* if $aRb \wedge bRc \Rightarrow aRc$ for all $a, b, c \in A$.
- *symmetric* if $aRb \Rightarrow bRa$ for all $a, b \in A$.
- *anti-symmetric* if $aRb \wedge bRa \Rightarrow a = b$ for all $a, b \in A$.
- an *equivalence relation* if it's reflexive, transitive, and symmetric.
- a *partial order* if it's reflexive, transitive, and anti-symmetric.
- a *strict partial order* if it's irreflexive, transitive, and anti-symmetric.

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Proof of $A_L = L$

To prove that $A_L = L$ we show for all $w \in \Sigma^*$ that

$$\hat{\delta}_L([\epsilon]_L, w) = [w]_L . \quad (1)$$

Then we conclude that

$$\begin{aligned} w \in L(A_L) &\Leftrightarrow \hat{\delta}_L([\epsilon]_L, w) \in F_L && \text{def } L(\text{DFA}) \\ &\Leftrightarrow [w]_L \in F_L && \text{by (1)} \\ &\Leftrightarrow w \in L \end{aligned}$$

Proof of $\hat{\delta}_L([\epsilon]_L, w) = [w]_L$

Proof.

by induction on the structure of w (Using appending construction - see Tutorial 1).

Base case: $\hat{\delta}_L([\epsilon]_L, \epsilon) = [\epsilon]_L$ by definition of $\hat{\delta}$ for DFAs.

Inductive case:

$$\begin{aligned}\hat{\delta}_L([\epsilon]_L, av) &= \hat{\delta}_L(\delta_L([\epsilon]_L, v), a) && \text{def. } \hat{\delta} \text{ for DFAs} \\ &= \delta_L([v]_L, a) && \text{ind. hyp.} \\ &= [va]_L && \text{def. } \delta_L \quad \square\end{aligned}$$